

| S                                         | F | D | F       | D | S | F                       | D | S | S       | F | D |
|-------------------------------------------|---|---|---------|---|---|-------------------------|---|---|---------|---|---|
| B                                         | D | C | D       | C | D | B                       | D | A | C       | D | C |
| A                                         | B | B | B       | D | B | C                       | A | B | B       | B | B |
| C                                         | A | A | A       | B | C | A                       | B | C | A       | C | A |
| D                                         | C | D | C       | A | A | D                       | C | D | D       | A | D |
| block 1                                   |   |   | block 2 |   |   | block 3                 |   |   | block 4 |   |   |
| whole plots: F flood, S sprinkler, D drip |   |   |         |   |   | subplots: varieties A-D |   |   |         |   |   |

Figure 33.1.1: A split-plot layout with  $r = 4$  blocks,  $a = 3$  irrigation regimes assigned at random within each block, and  $b = 4$  varieties assigned at random within each whole plot.

## 33.1 Split-plot designs

Some treatments cannot be applied to small units: an irrigation rig waters a strip of field, not a square metre of it; a teaching method is delivered to a class, not to a pupil. An experiment combining such a treatment with one applied as finely as one likes gives the coarse treatment to large units and the fine treatment to subdivisions of them — a **split-plot** design, named after the field trials it came from. Two independent randomizations mean two error terms, so that an analysis ignoring the distinction gets both wrong, in opposite directions.

### 33.1.1 The design and its randomization

Take  $r$  blocks. Each is divided into  $a$  **whole plots**, carrying the  $a$  levels of a factor  $A$  in random order: a randomized complete block design (Definition 18.4.1). Each whole plot is then divided into  $b$  **subplots**, carrying the  $b$  levels of a factor  $C$  in random order. The two randomizations are independent, and the second is repeated separately inside every whole plot. Figure 33.1.1 shows one such layout.

Two things vary from subplot to subplot: whatever makes whole plots differ, shared by all  $b$  subplots of a whole plot, and whatever makes subplots differ within a whole plot, which is not. So the model carries two error terms.

#### Definition 33.1.1. The balanced split-plot model

Index blocks by  $i = 1, \dots, r$ , whole-plot treatments by  $j = 1, \dots, a$  and subplot treatments by  $k = 1, \dots, b$ , and write  $y_{ijk}$  for the response on the subplot of block  $i$  receiving whole-plot treatment  $j$  and subplot treatment  $k$ . The **split-plot model** is

$$y_{ijk} = \mu + \xi_i + \omega_j + \eta_{ij} + \tau_k + \gamma_{jk} + e_{ijk}, \quad (33.1.1)$$

where  $\xi_i$ ,  $\omega_j$ ,  $\tau_k$  and  $\gamma_{jk}$  are fixed effects, the  $\eta_{ij}$  are uncorrelated **whole-plot errors** with mean zero and variance  $\sigma_w^2$ , the  $e_{ijk}$  are uncorrelated **subplot errors** with mean zero and variance  $\sigma_s^2$ , and the two sets are uncorrelated with each other. The **normal split-plot**

**model** takes both sets to be normal.

The whole plot  $(i, j)$  is a cluster of  $b$  observations that share  $\eta_{ij}$ . Writing  $\mathbf{Z}_w$  for the  $n \times ra$  matrix of whole-plot indicators, with  $n = rab$ , the combined error  $\eta_{ij} + e_{ijk}$  has

$$\text{Cov}(\mathbf{Y}) = \mathbf{V} = \sigma_s^2 \mathbf{I} + \sigma_w^2 \mathbf{Z}_w \mathbf{Z}_w^\top = \sigma_s^2 \mathbf{I} + b \sigma_w^2 \mathbf{P}_W, \quad (33.1.2)$$

where  $\mathbf{P}_W = b^{-1} \mathbf{Z}_w \mathbf{Z}_w^\top$  replaces every observation by its whole-plot mean (Example (The one-way layout)). Every pair of subplots in a whole plot has correlation  $\rho = \sigma_w^2 / (\sigma_w^2 + \sigma_s^2)$ , pairs in different whole plots none. The next section studies this structure in general; here balance makes everything explicit.

Where does the equicorrelation come from? In a randomized experiment, random assignment plus unit–treatment additivity *produces* the model rather than assuming it (Section 15.1), and here the argument runs twice: the second randomization is performed anew in every whole plot, making the subplots of a whole plot exchangeable — exactly the covariance Eq. 33.1.2. When the subplot factor is not randomized, as it is not when that factor is time (Section 33.3), the justification disappears and with it the equicorrelation.

### 33.1.2 Two strata

Order the observations with  $k$  fastest, then  $j$ , then  $i$ , and use the averaging and centring matrices  $\bar{\mathbf{J}}_l$ ,  $\mathbf{C}_l$  of Section 16.1. Define

$$\begin{aligned} \mathbf{P}_0 &= \bar{\mathbf{J}}_r \otimes \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b, & \mathbf{P}_B &= \mathbf{C}_r \otimes \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b, & \mathbf{P}_A &= \bar{\mathbf{J}}_r \otimes \mathbf{C}_a \otimes \bar{\mathbf{J}}_b, \\ \mathbf{P}_{AB} &= \mathbf{C}_r \otimes \mathbf{C}_a \otimes \bar{\mathbf{J}}_b, & \mathbf{P}_C &= \bar{\mathbf{J}}_r \otimes \bar{\mathbf{J}}_a \otimes \mathbf{C}_b, & \mathbf{P}_{AC} &= \bar{\mathbf{J}}_r \otimes \mathbf{C}_a \otimes \mathbf{C}_b, \end{aligned} \quad (33.1.3)$$

and  $\mathbf{P}_E = \mathbf{C}_r \otimes \mathbf{I}_a \otimes \mathbf{C}_b$ , the terms of the balanced three-way layout with one observation per cell (Theorem 16.3.1):  $\mathbf{P}_{AB}$  is the block-by- $A$  interaction, and  $\mathbf{P}_E$  collects the block-by- $C$  and three-factor interactions. The split-plot analysis rests on the fact that these seven projections fall into two groups according to how  $\mathbf{V}$  acts on them.

#### Lemma 33.1.1. The two error strata

Let  $\mathbf{V}$  be as in Eq. 33.1.2 and put  $\lambda_w = \sigma_s^2 + b \sigma_w^2$  and  $\lambda_s = \sigma_s^2$ . Then:

- (a)  $\mathbf{P}_W = \mathbf{I}_r \otimes \mathbf{I}_a \otimes \bar{\mathbf{J}}_b$ , and  $\mathbf{V} = \lambda_w \mathbf{P}_W + \lambda_s (\mathbf{I} - \mathbf{P}_W)$ , so  $\mathbf{V}$  has the two eigenvalues  $\lambda_w$  and  $\lambda_s$  with eigenspaces  $\mathcal{C}(\mathbf{P}_W)$  and  $\mathcal{C}(\mathbf{P}_W)^\perp$ , of dimensions  $ra$  and  $ra(b-1)$ .
- (b) The six matrices of Eq. 33.1.3 together with  $\mathbf{P}_E$  are mutually orthogonal projections summing to  $\mathbf{I}$ , with ranks 1,  $r-1$ ,  $a-1$ ,  $(r-1)(a-1)$ ,  $b-1$ ,  $(a-1)(b-1)$  and  $(r-1)a(b-1)$ .
- (c)  $\mathbf{P}_0, \mathbf{P}_B, \mathbf{P}_A, \mathbf{P}_{AB}$  have their ranges inside  $\mathcal{C}(\mathbf{P}_W)$ , and  $\mathbf{P}_C, \mathbf{P}_{AC}, \mathbf{P}_E$  have their ranges inside  $\mathcal{C}(\mathbf{P}_W)^\perp$ . Hence  $\mathbf{V}\mathbf{P} = \lambda_w \mathbf{P}$  for the first four and  $\mathbf{V}\mathbf{P} = \lambda_s \mathbf{P}$  for the last three.

*Proof.*

- (a) The whole plot  $(i, j)$  consists of the  $b$  observations with those first two indices, so averaging within whole plots is  $\mathbf{I}_r \otimes \mathbf{I}_a \otimes \bar{\mathbf{J}}_b$ , which is symmetric and idempotent of rank  $ra$  by Lemma 16.1.1(b). Substituting  $\mathbf{P}_W$  in Eq. 33.1.2 gives  $\mathbf{V} = \sigma_s^2 (\mathbf{I} - \mathbf{P}_W) + (\sigma_s^2 + b \sigma_w^2) \mathbf{P}_W$ .

- (b) Ranks, symmetry and idempotence are Lemma 16.1.1(b), noting  $\mathbf{P}_E = \mathbf{C}_r \otimes \bar{\mathbf{J}}_a \otimes \mathbf{C}_b + \mathbf{C}_r \otimes \mathbf{C}_a \otimes \mathbf{C}_b$ . Distinct terms differ in at least one slot, so they are orthogonal by Lemma 16.1.1(c), and they are the  $2^3$  products of Lemma 16.1.1(d) regrouped, so they sum to  $\mathbf{I}$ .
- (c) A product  $\mathbf{E}_1 \otimes \mathbf{E}_2 \otimes \bar{\mathbf{J}}_b$  satisfies  $\mathbf{P}_W \mathbf{P} = \mathbf{P}$  by the mixed-product rule and  $\bar{\mathbf{J}}_b \bar{\mathbf{J}}_b = \bar{\mathbf{J}}_b$ ; a product  $\mathbf{E}_1 \otimes \mathbf{E}_2 \otimes \mathbf{C}_b$  satisfies  $\mathbf{P}_W \mathbf{P} = \mathbf{0}$  because  $\bar{\mathbf{J}}_b \mathbf{C}_b = \mathbf{0}$ . The first four matrices are of the first kind and the last three of the second. Then  $\mathbf{V} \mathbf{P} = \lambda_w \mathbf{P}_W \mathbf{P} + \lambda_s (\mathbf{P} - \mathbf{P}_W \mathbf{P})$  gives the two cases. ■

Part (c) is the whole idea. The **whole-plot stratum**  $\mathcal{C}(\mathbf{P}_W)$  carries the whole-plot error intact, so its variance is  $\lambda_w$ ; in the **subplot stratum**  $\mathcal{C}(\mathbf{P}_W)^\perp$  that error is differenced away and only  $\sigma_s^2$  remains.

*Idea.* A split-plot design is two experiments sharing one field: a whole-plot experiment with  $ra$  units and error variance  $\sigma_s^2 + b\sigma_w^2$ , and a subplot experiment with  $rab$  units and error variance  $\sigma_s^2$ . Effects constant within whole plots belong to the first and effects averaging to zero within every whole plot to the second — the interaction among them, which is why such a design studies an interaction well and the coarse main effect badly.

### 33.1.3 The analysis

#### Theorem 33.1.2. The balanced split-plot design

Assume Eq. 33.1.1 with  $r, a, b \geq 2$ , and write the cell means  $\mu_{ijk} = E(y_{ijk})$ . Let  $\mathbf{M}$  be the orthogonal projection onto the model space  $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{P}_0 + \mathbf{P}_B + \mathbf{P}_A + \mathbf{P}_C + \mathbf{P}_{AC})$ .

- (a)  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , so least squares is generalized least squares and every least squares estimate of an estimable function is its best linear unbiased estimator, whatever  $\sigma_w^2$  and  $\sigma_s^2$  are.
- (b) The sums of squares are those of the balanced three-way layout, with  $SS_A = rb \sum_j (\bar{y}_{\cdot j} - \bar{y}_{\dots})^2$ ,  $SS_C = ra \sum_k (\bar{y}_{\cdot k} - \bar{y}_{\dots})^2$ ,  $SS_{AC} = r \sum_{j,k} (\bar{y}_{jk} - \bar{y}_{\cdot j} - \bar{y}_{\cdot k} + \bar{y}_{\dots})^2$ , and the two error sums of squares  $SS_{AB} = \|\mathbf{P}_{AB}\mathbf{y}\|^2$  and  $SSE = \|\mathbf{P}_E\mathbf{y}\|^2$ .
- (c) The expected mean squares (Theorem 9.6.1) are

$$\begin{aligned} E \text{ MS}_B &= \lambda_w + \frac{ab}{r-1} \sum_i (\xi_i - \bar{\xi})^2, & E \text{ MS}_{AB} &= \lambda_w = \sigma_s^2 + b\sigma_w^2, \\ E \text{ MS}_A &= \lambda_w + \frac{rb}{a-1} \sum_j (\alpha_j - \bar{\alpha})^2, & E \text{ MSE} &= \lambda_s = \sigma_s^2, \\ E \text{ MS}_C &= \lambda_s + \frac{ra}{b-1} \sum_k (\kappa_k - \bar{\kappa})^2, & E \text{ MS}_{AC} &= \lambda_s + \frac{r}{(a-1)(b-1)} \sum_{j,k} \delta_{jk}^2, \end{aligned} \quad (33.1.4)$$

where  $\alpha_j = \omega_j + \bar{\gamma}_{\cdot j}$ ,  $\kappa_k = \tau_k + \bar{\gamma}_{\cdot k}$  and  $\delta_{jk} = \gamma_{jk} - \bar{\gamma}_{j\cdot} - \bar{\gamma}_{\cdot k} + \bar{\gamma}$  are the equal-weight effects of Definition 16.2.1 computed from  $\bar{\mu}_{\cdot jk}$ .

- (d) Under normality the six sums of squares are independent,  $SS_{AB}/\lambda_w \sim \chi^2((r-1)(a-1))$ ,

$\text{SSE}/\lambda_s \sim \chi^2((r-1)a(b-1))$ , and

$$\frac{\text{MS}_A}{\text{MS}_{AB}} \sim F\left(a-1, (r-1)(a-1), \frac{rb}{\lambda_w} \sum_j (\alpha_j - \bar{\alpha})^2\right),$$

while  $\text{MS}_C/\text{MSE}$  and  $\text{MS}_{AC}/\text{MSE}$  have noncentral  $F$  distributions on  $(b-1, (r-1)a(b-1))$  and  $((a-1)(b-1), (r-1)a(b-1))$  degrees of freedom with noncentralities  $ra \sum_k (\kappa_k - \bar{\kappa})^2/\lambda_s$  and  $r \sum_{j,k} \delta_{jk}^2/\lambda_s$ . So the whole-plot factor is tested against  $\text{MS}_{AB}$  and the subplot factor and the interaction against  $\text{MSE}$ .

- (e) A contrast  $\sum_j c_j \alpha_j$  is estimated by  $\sum_j c_j \bar{y}_{.j}$  with variance  $\lambda_w \sum_j c_j^2/(rb)$ ; a contrast  $\sum_k d_k \kappa_k$  is estimated by  $\sum_k d_k \bar{y}_{.k}$  with variance  $\lambda_s \sum_k d_k^2/(ra)$ . Both are best linear unbiased, and the two estimators are uncorrelated.
- (f) Fix a subplot treatment  $k$  and a contrast  $\sum_j c_j = 0$ . The simple effect  $\sum_j c_j \bar{\mu}_{.jk}$  is estimated by  $\sum_j c_j \bar{y}_{.jk}$ , best linear unbiased, with variance

$$\left(\sum_j c_j^2\right) \frac{\lambda_w + (b-1)\lambda_s}{rb} = \left(\sum_j c_j^2\right) \frac{\sigma_w^2 + \sigma_s^2}{r}, \quad (33.1.5)$$

which belongs to neither stratum: no single mean square estimates it, and the unbiased estimate is the combination  $(\sum_j c_j^2)\{\text{MS}_{AB} + (b-1)\text{MSE}\}/(rb)$ .

*Proof.*

- (a) Every  $\mathbf{v} \in \mathcal{C}(\mathbf{X})$  splits as  $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$  with  $\mathbf{v}_1 \in \mathcal{C}(\mathbf{P}_0 + \mathbf{P}_B + \mathbf{P}_A)$  and  $\mathbf{v}_2 \in \mathcal{C}(\mathbf{P}_C + \mathbf{P}_{AC})$ , because the five projections are mutually orthogonal and sum to  $\mathbf{M}$ . By Lemma 33.1.1(c),  $\mathbf{V}\mathbf{v} = \lambda_w \mathbf{v}_1 + \lambda_s \mathbf{v}_2$ , again in  $\mathcal{C}(\mathbf{X})$ . So  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , and Theorem 6.9.3 says the ordinary and the  $\mathbf{V}^{-1}$  projection onto  $\mathcal{C}(\mathbf{X})$  coincide. The estimator is therefore both the least squares and the generalized least squares estimator, which is the best linear unbiased estimator by Corollary 7.2.3. (This is the balanced case of Theorem 31.2.2.)
- (b) Each projection averages over the slots carrying  $\bar{\mathbf{J}}$  and centres in the slots carrying  $\mathbf{C}$ . For instance  $\mathbf{P}_A = \bar{\mathbf{J}}_r \otimes \mathbf{I}_a \otimes \bar{\mathbf{J}}_b - \bar{\mathbf{J}}_r \otimes \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$  puts  $\bar{y}_{.j} - \bar{y}_{...}$  in position  $(i, j, k)$ , a value repeated  $rb$  times, giving  $\text{SS}_A$ ; the same expansion gives  $\text{SS}_C$  and  $\text{SS}_{AC}$ , as in Theorem 16.3.1(b). The error sums of squares are the squared lengths of the remaining components.
- (c) Each  $\mathbf{P}$  in the list satisfies  $\mathbf{V}\mathbf{P} = \lambda\mathbf{P}$  with  $\lambda$  the eigenvalue of its stratum, so  $\text{E}(\mathbf{Y}^\top \mathbf{P}\mathbf{Y}) = \text{tr}(\mathbf{P}\mathbf{V}) + \boldsymbol{\mu}^\top \mathbf{P}\boldsymbol{\mu} = \lambda \text{rank}(\mathbf{P}) + \|\mathbf{P}\boldsymbol{\mu}\|^2$  by Theorem 2.3.1, where  $\boldsymbol{\mu}$  is the mean vector. Dividing by the rank gives  $\text{E}(\text{MS}) = \lambda + \|\mathbf{P}\boldsymbol{\mu}\|^2/\text{rank}(\mathbf{P})$ . It remains to evaluate  $\mathbf{P}\boldsymbol{\mu}$ . Since  $\boldsymbol{\mu}$  has entries  $\mu + \xi_i + \omega_j + \tau_k + \gamma_{jk}$ , the projection  $\mathbf{P}_A$  returns  $\bar{\mu}_{.j} - \bar{\mu}_{...} = \alpha_j - \bar{\alpha}$  on each of the  $rb$  observations with whole-plot treatment  $j$ ; likewise  $\mathbf{P}_C$  returns  $\kappa_k - \bar{\kappa}$  on each of  $ra$  observations,  $\mathbf{P}_{AC}$  returns  $\delta_{jk}$  on each of  $r$ , and  $\mathbf{P}_B$  returns  $\xi_i - \bar{\xi}$  on each of  $ab$ . Finally  $\mathbf{P}_{AB}\boldsymbol{\mu} = \mathbf{0}$  and  $\mathbf{P}_E\boldsymbol{\mu} = \mathbf{0}$ , because  $\boldsymbol{\mu}$  has no block-by-treatment terms.
- (d) Write  $\mathbf{V}^{1/2} = \lambda_w^{1/2} \mathbf{P}_W + \lambda_s^{1/2} (\mathbf{I} - \mathbf{P}_W)$ , a symmetric positive definite square root (Theorem 1.7.6), and put  $\mathbf{z} = \mathbf{V}^{-1/2} \mathbf{Y} \sim \text{N}_n(\mathbf{V}^{-1/2} \boldsymbol{\mu}, \mathbf{I})$  (Proposition 2.2.4). For any  $\mathbf{P}$  in the list, Lemma 33.1.1(c) gives  $\mathbf{V}^{1/2} \mathbf{P} \mathbf{V}^{1/2} = \lambda \mathbf{P}$ , so  $\mathbf{Y}^\top \mathbf{P} \mathbf{Y} = \lambda \mathbf{z}^\top \mathbf{P} \mathbf{z}$ . The projections are mutually orthogonal, so Theorem 4.4.4 applied to  $\mathbf{z}$  makes the quadratic forms  $\mathbf{z}^\top \mathbf{P} \mathbf{z}$  independent with noncentral chi-squared distributions on  $\text{rank}(\mathbf{P})$  degrees of freedom and noncentrality  $\|\mathbf{P}\mathbf{V}^{-1/2} \boldsymbol{\mu}\|^2 = \|\mathbf{P}\boldsymbol{\mu}\|^2/\lambda$ , the last equality because  $\mathbf{V}^{-1/2}$  acts on  $\mathcal{C}(\mathbf{P})$  as multiplication by  $\lambda^{-1/2}$ . Forming ratios and cancelling  $\lambda$  be-

tween numerator and denominator *within the same stratum* gives the stated  $F$  distributions (Definition 4.1.2). Ratios across strata would not cancel, which is precisely why  $MS_A/MSE$  is not an  $F$  statistic.

- (e) The coefficient vector of  $\sum_j c_j \bar{y}_{.j}$  is  $\boldsymbol{\rho} = (rb)^{-1} \mathbf{1}_r \otimes \mathbf{c} \otimes \mathbf{1}_b \in \mathcal{C}(\mathbf{P}_A)$ , since  $\mathbf{c} \perp \mathbf{1}_a$ , so by Lemma 33.1.1(c)  $\text{Var}(\boldsymbol{\rho}^\top \mathbf{Y}) = \lambda_w \|\boldsymbol{\rho}\|^2 = \lambda_w \sum_j c_j^2 / (rb)$ . The same computation with  $\boldsymbol{\rho}_C = (ra)^{-1} \mathbf{1}_r \otimes \mathbf{1}_a \otimes \mathbf{d} \in \mathcal{C}(\mathbf{P}_C)$  gives the second variance, and the two are uncorrelated because  $\boldsymbol{\rho}^\top \boldsymbol{\rho}_C = 0$ . Both are least squares estimates of estimable functions, hence best linear unbiased by (a).
- (f) Split  $\mathbf{e}_k = b^{-1} \mathbf{1}_b + (\mathbf{e}_k - b^{-1} \mathbf{1}_b)$ . The coefficient vector  $r^{-1} \mathbf{1}_r \otimes \mathbf{c} \otimes \mathbf{e}_k$  splits accordingly into  $\boldsymbol{\rho}_1 \in \mathcal{C}(\mathbf{P}_A)$  and  $\boldsymbol{\rho}_2 \in \mathcal{C}(\mathbf{P}_{AC})$ , of squared lengths  $\sum_j c_j^2 / (rb)$  and  $(b-1) \sum_j c_j^2 / (rb)$ . They lie in different strata and are orthogonal, so the variance is  $\lambda_w \|\boldsymbol{\rho}_1\|^2 + \lambda_s \|\boldsymbol{\rho}_2\|^2$ , which is Eq. 33.1.5; the estimate of it is unbiased by Eq. 33.1.4, and the estimator is best linear unbiased by (a). ■

Dropping  $(\xi\omega)_{ij}$  from a full three-way model is what turns the block-by- $A$  interaction into the error term for  $A$ , on only  $(r-1)(a-1)$  degrees of freedom: the formal content of “whole plots are the experimental units for  $A$ ”. And  $MS_A$  tests a contrast in  $\alpha_j = \omega_j + \bar{\gamma}_{j.}$ , the whole-plot effect averaged over the subplot treatments used, not in  $\omega_j$  alone (Section 16.2).

Part (f) is the case most often got wrong. “The interaction is tested against MSE” tempts one to compare two whole-plot treatments *within a single subplot treatment* against MSE too; but that comparison carries the whole-plot error intact, and Eq. 33.1.5 prices it between the two strata. Its estimated variance combines two mean squares, so no exact  $t$  or  $F$  test exists, and Satterthwaite’s rule (Proposition 32.5.1) gives it

$$\nu = \frac{\{MS_{AB} + (b-1)MSE\}^2}{MS_{AB}^2 / \{(r-1)(a-1)\} + (b-1)^2 MSE^2 / \{(r-1)a(b-1)\}} \quad (33.1.6)$$

approximate degrees of freedom, which lie between the two exact values.

### Example. An irrigation and variety trial

Four fields are each divided into three whole plots, and the regimes flood, sprinkler and drip are assigned at random within each field; each whole plot is divided into four subplots carrying four barley varieties in random order, giving  $n = 48$  yields in tonnes per hectare, simulated from Eq. 33.1.1 with  $\sigma_w = 0.35$  and  $\sigma_s = 0.25$  t/ha. The analysis of variance is

| Source                      | df | SS    | MS    | $F$    | $p$    |
|-----------------------------|----|-------|-------|--------|--------|
| blocks                      | 3  | 9.471 | 3.157 |        |        |
| irrigation $A$              | 2  | 9.215 | 4.608 | 10.637 | 0.0106 |
| whole-plot error            | 6  | 2.599 | 0.433 |        |        |
| variety $C$                 | 3  | 1.755 | 0.585 | 9.442  | 0.0002 |
| irrigation $\times$ variety | 6  | 0.958 | 0.160 | 2.577  | 0.0419 |
| subplot error               | 27 | 1.673 | 0.062 |        |        |

The two error mean squares differ by a factor of seven. Solving Eq. 33.1.4 gives  $\hat{\sigma}_s = \sqrt{\text{MSE}} = 0.2489$  and  $\hat{\sigma}_w = \sqrt{(\text{MS}_{AB} - \text{MSE})/b} = 0.3046$ , close to the generating values.

The irrigation means are 4.8075, 5.8000 and 5.6575 t/ha; a difference of two has standard error  $\sqrt{2\text{MS}_{AB}/(rb)} = 0.2327$  on 6 degrees of freedom, so drip beats flood by 0.8500 t/ha with  $t = 3.653$ . The variety means are 5.134, 5.364, 5.574 and 5.614, and a difference of two has standard error  $\sqrt{2\text{MSE}/(ra)} = 0.1016$  on 27 degrees of freedom — more than twice as precise, from the same 48 observations. Comparing drip with flood *for one variety* is neither: by Eq. 33.1.5 its standard error is  $\sqrt{\{\text{MS}_{AB} + 3\text{MSE}\}/8} = 0.2782$ , on 11.77 Satterthwaite degrees of freedom.

```
import numpy as np

r, a, b = 4, 3, 4                                # blocks, irrigation regimes,
    varieties
n = r * a * b
block_eff = np.array([-0.30, 0.10, 0.05, 0.15])   # field-to-field fertility
irrig_eff = np.array([0.00, 0.45, 0.70])          # flood, sprinkler, drip
variety_eff = np.array([-0.30, -0.10, 0.15, 0.25])
inter = np.array([[0.10, -0.05, -0.10, 0.05],     # irrigation x variety
                  [-0.05, 0.10, 0.00, -0.05],
                  [-0.05, -0.05, 0.10, 0.00]])
sigma_w, sigma_s = 0.35, 0.25                    # whole-plot and subplot errors

rng = np.random.default_rng(3301)
eta = rng.normal(0, sigma_w, (r, a))              # one error per whole plot
e = rng.normal(0, sigma_s, (r, a, b))             # one error per subplot
mean = (5.0 + block_eff[:, None, None] + irrig_eff[None, :, None]
        + variety_eff[None, None, :] + inter[None, :, :])
Y = np.round(mean + eta[:, :, None] + e, 2)       # yields, t/ha
y = Y.reshape(-1)                                # stacked block, then regime,
    then variety
```

[Run this code online](#)

The analysis is seven inner products, each projection a Kronecker product as in Section 16.3, with two denominators in place of one.

```
from scipy import stats

def Jbar(k):
    return np.full((k, k), 1.0 / k)

def Cen(k):
    return np.eye(k) - Jbar(k)

def kron3(E1, E2, E3):
    return np.kron(np.kron(E1, E2), E3)

P = {"blocks": kron3(Cen(r), Jbar(a), Jbar(b)),
     "irrigation": kron3(Jbar(r), Cen(a), Jbar(b)),
     "whole-plot error": kron3(Cen(r), Cen(a), Jbar(b)),
     "varieties": kron3(Jbar(r), Jbar(a), Cen(b)),
     "irrigation x variety": kron3(Jbar(r), Cen(a), Cen(b)),
     "subplot error": kron3(Cen(r), np.eye(a), Cen(b))}
df = {k: int(round(np.trace(M))) for k, M in P.items()}
SS = {k: y @ M @ y for k, M in P.items()}
```

```

MS = {k: SS[k] / df[k] for k in P}

F_irrigation = MS["irrigation"] / MS["whole-plot error"]
F_variety = MS["varieties"] / MS["subplot error"]
F_inter = MS["irrigation x variety"] / MS["subplot error"]
for name, F, d in [("irrigation", F_irrigation, df["whole-plot error"]),
                  ("varieties", F_variety, df["subplot error"]),
                  ("irrigation x variety", F_inter, df["subplot error"])):
    p = stats.f.sf(F, df[name], d)
    print(f"{name:22s} F({df[name]}, {d}) = {F:7.3f}    p = {p:.4f}")

```

[Run this code online](#)

### 33.1.4 What pooling costs

Pool the two error sums of squares into one residual mean square on 33 degrees of freedom — in [Example \(An irrigation and variety trial\)](#), 0.129 — and by [Eq. 33.1.4](#) the consequences are opposite in the two strata. The whole-plot denominator falls towards  $\lambda_s$ , inflating the irrigation  $F$  to 35.591 on (2, 33) degrees of freedom,  $p = 5.78 \times 10^{-9}$  against 0.0106, and shrinking the standard error of an irrigation difference from 0.2327 to 0.1272, a factor of 1.829 too small; the subplot denominator exceeds  $\lambda_s$ , and that of a variety difference rises from 0.1016 to 0.1469, a factor of 1.445 too large. Pooling manufactures significance for the treatment the design studies badly and destroys it for the one it studies well.

*Warning.* The two error terms are a property of the *randomization*, not of the data. A split-plot experiment analysed as a completely randomized three-factor experiment produces a plausible table with one residual line, and nothing in the numbers announces the mistake. Wherever one factor is applied to groups of units and another within them, look for two errors.

### 33.1.5 The same analysis as a mixed model

[Eq. 33.1.1](#) is a linear mixed model (Definition 32.2.1) with  $\mathbf{Z} = \mathbf{Z}_w$ ,  $\mathbf{G} = \sigma_w^2 \mathbf{I}_{ra}$  and  $\mathbf{R} = \sigma_s^2 \mathbf{I}_n$ , whose marginal covariance  $\mathbf{ZGZ}^\top + \mathbf{R}$  is [Eq. 33.1.2](#). In the balanced case, fitting it by restricted maximum likelihood (Definition 32.4.1) reproduces the analysis above exactly, provided  $\text{MS}_{AB} \geq \text{MSE}$ : the restricted likelihood depends on the data only through the two mean squares, and is maximized at the moment estimates of [Example \(An irrigation and variety trial\)](#) ([Exercise \(B2\)](#)). Satterthwaite's rule (Proposition 32.5.1) then returns exactly  $(r-1)(a-1)$  degrees of freedom for a whole-plot contrast, and [Eq. 33.1.6](#) for a simple effect.

The route earns its keep when the balance fails: a missing subplot, a whole plot lost to flooding, a covariate measured at the subplot level. The strata are then no longer orthogonal, no mean square estimates  $\lambda_w$  or  $\lambda_s$  on its own, and Henderson's equations (Theorem 32.3.2) do the computing. Two variations change nothing essential: the whole-plot design may be completely randomized or a Latin square (Definition 18.5.1), which changes only the whole-plot degrees of freedom, and splitting can be repeated, giving three strata ([Exercise \(C1\)](#)).

### 33.1.6 Exercises

#### 33.1.6.1 A. Check your understanding

**Exercise. A1**

For the design of [Example \(An irrigation and variety trial\)](#), say which stratum each of the following belongs to, and on how many degrees of freedom its error is estimated: the difference between two block means; between two variety means; between two irrigation means; between two varieties within the drip regime; the difference in the effect of drip versus flood between two varieties; and the difference between drip and flood for the first variety alone.

*Solution.* The block and irrigation differences are constant within whole plots, so they lie in the whole-plot stratum and use  $MS_{AB}$  on 6 degrees of freedom. The next three are contrasts summing to zero within each whole plot, so they lie in the subplot stratum and use MSE on 27 degrees of freedom. The last lies in neither: it is [Theorem 33.1.2\(f\)](#), with estimated variance  $\{MS_{AB} + 3MSE\}/8$  on about 11.8 Satterthwaite degrees of freedom.

**Exercise. A2**

With  $r$ ,  $a$ ,  $\sigma_w^2$  and  $\sigma_s^2$  fixed, what happens to the variance of a whole-plot contrast, and to that of a subplot contrast, as  $b$  grows? Which kind of comparison does adding subplots help?

**33.1.6.2 B. Practice****Exercise. B1**

Verify [Theorem 33.1.2\(a\)](#) directly from [Theorem 6.9.3](#) by computing  $\mathbf{V}\mathbf{X}$  for the model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{B}, \mathbf{A}, \mathbf{C}, \mathbf{AC}]$  of indicator columns. Then show that adding the block-by- $A$  columns leaves the condition true but destroys the analysis, and say why.

**Exercise. B2**

Show that for the balanced split-plot model the restricted likelihood of  $(\sigma_w^2, \sigma_s^2)$  depends on the data only through  $SS_{AB}$  and SSE, and that it is maximized at  $\hat{\lambda}_w = MS_{AB}$ ,  $\hat{\lambda}_s = MSE$  provided  $MS_{AB} \geq MSE$ . *Hint:* the error contrasts span  $\mathcal{C}(\mathbf{P}_{AB}) \oplus \mathcal{C}(\mathbf{P}_E)$ , and  $\mathbf{V}$  acts as a scalar on each.

*Solution.* The error contrasts are the  $\mathbf{a}^\top \mathbf{y}$  with  $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{P}_{AB}) \oplus \mathcal{C}(\mathbf{P}_E)$ . Take orthonormal bases of the two summands, of sizes  $f_w = (r-1)(a-1)$  and  $f_s = (r-1)a(b-1)$ . By [Lemma 33.1.1\(c\)](#) the transformed data are independent,  $N(0, \lambda_w)$  for the first basis and  $N(0, \lambda_s)$  for the second. By [Definition 32.4.1](#) the restricted log likelihood is therefore  $-\frac{1}{2}\{f_w \log \lambda_w + SS_{AB}/\lambda_w + f_s \log \lambda_s + SSE/\lambda_s\}$  up to a constant, a sum of two ordinary normal-variance likelihoods maximized at the two mean squares. Since  $\sigma_s^2 = \lambda_s$  and  $\sigma_w^2 = (\lambda_w - \lambda_s)/b$ , the constraint  $\sigma_w^2 \geq 0$  is  $MS_{AB} \geq MSE$ .

**33.1.6.3 C. Going deeper****Exercise. C1**

A split-split-plot design divides each subplot into  $c$  sub-subplots and randomizes a third factor  $D$  among them, adding an error  $\zeta_{ijk}$  of variance  $\sigma_u^2$  shared by the sub-subplots of a subplot. Write down the covariance matrix in the style of Eq. 33.1.2, find its three eigenvalues and eigenspaces, and state which effects are tested against which of the three error mean squares.